MH2500 Probability and Introduction to Statistics

Slides for Week 1

Topics: Sample Spaces, Events and Counting Principles

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Welcome to MH2500!

- Probability theory is the branch of mathematics concerned with probability, which is based on courses: calculus, discrete mathematics, and linear algebra.
- It originated in the study of gambling games in 17-th century.

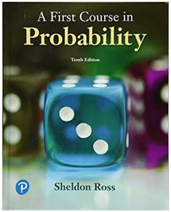
- Applications extended to
  - Insurance, risk assessment, stock market
  - Medical tests
  - Reliability
  - Resource allocation (e.g. airline seat inventory control)

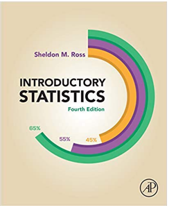
- It is a prerequisite of many courses in mathematics and statistics.

- After taking this course, you will understand why gamblers cannot win.

Textbooks by Sheldon Ross:

- Probability: *A First Course in Probability*, Pearson.
- Statistics: *Introductory Statistics* (Chapters 7 and 9), Academic Press.





Probability:

- Classical probability (Chapters 1–3)
- Discrete distributions (Chapter 4)
- Continuous distributions (Chapter 5)
- Joint distributions (Chapter 6)
- Covariance/Moment Generating Functions (Chapter 7)
- Limit Theorems (Chapter 8)

Statistics:

- Hypothesis Testing (Chapters 7 and 9)

- Lecturer: Wu Guohua, SPMS-MAS-05-40
- Time:
  - Lectures: Tuesday 09:30–10:30, via Zoom
    Friday 08:30–10:30, via Zoom
  - Tutorials: from week 2
    - *Tutorial questions will be uploaded on NTULearn weekly, and you will have enough time to work through these questions before you attend tutorials*
    - *Tutorial solution will be uploaded after tutorial sessions*
    - *Tutorial sessions will be used to discuss tutorial problems, clarify doubts, check your work, etc.*

Office Hour: Tuesday afternoon: 14:30-15:30

Course Evaluation

- Components of evaluation:
  - 20% Test 1 (50 minutes)
  - 20% Test 2 (50 minutes)
  - 10% Class Participation in Tutorials
  - 50% Final exam (2 hours)
- Test 1 on Tuesday, 21 September (Week 7)

  1 piece of A4 paper as helpsheet allowed.
- Test 2 on Tuesday, 26 October (Week 11)

  1 piece of A4 paper as helpsheet allowed.
- Final Exam

  1 piece of A4 paper as helpsheet allowed.

IMPORTANT: Absence from Tests

- Students who are absent from any Test without valid Leave of Absence will be given zero mark for the missed test
- If you have a valid Leave of Absence for any Test, please inform the Lecturer and submit official form for leave of absence to your school within 24 hours
- There will be NO replacement tests. The percentage will be shifted to other parts of the assessment if you miss a test with valid reason

Advice:

- Knowledge and Skills from Calculus, Discrete Mathematics and Linear Algebra, will be needed for this course
- Read textbooks, more practice, understand examples
- Attend tutorials, understand the idea of solving tutorial questions, and work out answers to assignment questions independently
- Study group, discussion
- Office Hours, Appointment with me and tutors

Try this question:

The digits 1, 2, 3, 4 and 5 are arranged randomly for form a five-digit number. *No digit is repeated*.

- Find the probability that the number is greater than 30 000.

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Don't worry if you find your idea is different from your friends.

Statistical Experiments and Observations

A statistical experiment is a process that generates a set of data.

Examples:

- flipping a coin
- tossing a die
- collecting votes

An observation is a recording of an experiment, numerical or categorical.

- Outcomes of flipping a coin: Head or Tail, are observations.
- Opinions of voters concerning a new sales tax are observations.

Sample Spaces

We are interested in the observations obtained by repeating a experiment several times.

- The outcomes will depend on chance only and cannot be predicted with certainty.

Definition:

The set of all possible outcomes of a statistical experiment is called the sample space and is represented by the symbol S, or sometimes Ω.

Each outcome in a sample space is called an element, or a member of the sample space, or a sample point.

Example 1: Flipping a Coin

When a unbiased coin is flipped, possible outcomes are Head (H) or Tail (T).

- These two outcomes are equally likely to occur.

The sample space S is

$$S = \{H, T\}$$

Note:

We list the elements in the sample space, when possible.

Example 2: Tossing a Die

When a die is tossed, possible outcomes are $1, 2, 3, 4, 5$ and 6.

The sample space S is

$$S = \{1, 2, 3, 4, 5, 6\}.$$

In the experiment of tossing two dice, the sample space consists of 36 sample points:

$$S = \{(i, j) : i, j = 1, 2, \cdots, 6\},$$

where the outcome (i, j) is said to occur if i appears on the left die and j on the right die.

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. . . .

In this experiment, the sample space is finite, 36 sample points only.

Exercise:

An experiment consists of flipping an unbiased coin and then

- flipping it a second time, if a head occurs;
- tossing a die, if a tail occurs.

List all possible outcomes of this sample space.

When a sample space is large or infinite, we may not be able to list all elements.

We use sets to denote sample spaces.

- $S = \{x \mid x \text{ is a city with population over 1 million}\}$

Infinite Sample Spaces

Example:

Throwing a dart on a circular board with radius 10 cm.

The outcome is the position of the dart from the center of the circle.

- Recall the equation of the circle is $x^2 + y^2 = 10^2$.

The sample space is

$$S = \left\{(x, y) \mid \; x^2 + y^2 \leq 10^2\right\}.$$

Example:

If the experiment consists of measuring (in hours) the lifetime of a transistor, the sample space consists of all nonnegative real numbers; that is $S = \{x : 0 \leq x < \infty\}$.

Events

An event is just a subset of the sample space, and we use upper-case letters to denote events: A, B, E, etc.

- An event can be the null set $\emptyset$, or the whole sample space S.

Roll two dice and add the sum of the two numbers on the dice. The sample space is

$$\Omega = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

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Let A be the event that an odd number is obtained. Then
$A = \{3, 5, 7, 9, 11\}$.
We can also have other events:
$B = \{2, 4, 6, 8, 10, 12\}$ and
$C = \{2, 3, 5, 7, 11\}$.

Complement, union and intersection

(a) The complement of an event A with respect to S is the event of all elements of S not in A — denote it by A'.

We also use A^c to denote the complement of A.

(b) The intersection of two events A and B is the event containing all elements that are common to A and B — denote it by $A \cap B$.

(c) The union of two events A and B is the event containing all elements that belong to A or B or both — denote it by $A \cup B$.

Exercise:

Tossing a die, and the sample space $S = \{1, 2, 3, 4, 5, 6\}$. Let

$$A = \{3, 6\}, B = \{1, 3, 5\}.$$

What are A', $A \cap B$ and $A \cup B$?

Useful Identities of Sets

Let S be a sample space, and A and B be events.

1. $\emptyset' = S$, $S' = \emptyset$, $(A')' = A$;
2. $A \cap A' = \emptyset$, $A \cup A' = S$;
3. $A \cap \emptyset = \emptyset$, $A \cup \emptyset = A$;
4. $(A \cap B)' = A' \cup B'$, $(A \cup B)' = A' \cap B'$.

The event $A \backslash B$ is the event containing elements in A but not in B.

$$A \backslash B = A \cap B'.$$

Example:

Let S be the patients in Jurong Clinic, A be the event that a patient has diabetes, and B be the event that a patient has high blood pressure.

Then the event $A \backslash B$ is the event that a patient has diabetes but not high blood pressure.

Mutually Exclusive Events

Two events A and B are mutually exclusive, or disjoint, if $A \cap B = \emptyset$, i.e., they do not have common elements.

Deck of Cards:

Suppose a card is selected at random from an ordinary deck of 52 playing cards.

Consider the event A that the card is red, B that the card is a club, and C that the card is an ACE.

- Events A and B are mutually exclusive.
- Events A and C are not mutually exclusive.

Counting Sample Points

We have learnt counting techniques from “Discrete Mathematics" (MH1301):

Addition Principle

If event A can occur in m ways, and event B can occur in n ways (disjoint from the m ways for event A), then the event “A or B” can occur in $m + n$ ways.

Example:

Tossing two dice and add the sum of the two numbers on the dice. Find the number of ways to obtain 4 as the sum.

Multiplication Principle

If two experiments are to be performed, and experiment 1 can result in any one of m possible outcomes, and for each outcome of experiment 1, there are n possible outcomes of experiment 2. Then all together, there are mn possible outcomes of the two experiments.

Example:

Tossing two dice. The sample space consists of 36 samples points:

$$(i, j) : i, j = 1, 2, \cdots, 6.$$

We can name these two dice as die 1 and die 2. Tossing die 1, we have 6 outcomes, and for each of these outcome, we toss die 2, and we again have 6 outcomes.

Extended Multiplication Principle

If there are p experiments with the first experiment having n_1 outcomes, the second experiment having n_2 outcomes, $\cdots$, the p-th experiment having n_p outcomes, then there are $n_1 n_2 \cdots n_p$ possible outcomes for the p experiments.

Example:

Sam is going to assemble a computer by himself. He has the choice of chips from two brands, a hard drive from four, memory from three, and an accessory bundle from five local stores. How many different ways can Sam order the parts?

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(Answer: $2 \times 4 \times 3 \times 5 = 120$)

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(Answer: $26^3 \cdot 10^4 = 175,760,000$)

Permutation

A permutation refers to an arrangement of all or part of a set of objects.

- Different order of objects is considered a different permutation.

Recall that for any non-negative integer n, the integer $n!$ (n factorial) is defined as

$$n! = n(n-1)(n-2)\cdots(2)(1)$$

with special case $0! = 1$.

Theorem

1. The number of permutations of n distinct objects is $n!$.
2. The number of permutations of n distinct objects taken r at a time is
$$_nP_r = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!}.$$

Exercise:

MH4200 consists of 6 boys and 4 girls. An examination is given, and the student are ranked according to their performance. Assume that no two students obtain the same score.

1. How many different rankings are possible?

2. If boys are ranked just among themselves and girls just among themselves, how many different rankings are possible?

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(Answer: $10!$; $(6!)(4!)$)

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Cyclic Arrangements

The number of permutations of n distinct objects arranged in a circle is $(n-1)!$.

Prove it.

Arrangements with repetition

The number of distinct permutations of n objects, of which n_1 are of one kind, n_2 of a second, ..., n_k of a k-th kind, is

$$\frac{n!}{n_1!n_2!\cdots n_k!},$$

denoted as

$$\binom{n}{n_1, n_2, \ldots, n_k}.$$

Exercise:

In how many ways can 7 graduate students be assigned to 1 triple and 2 double hotel rooms during a conference?

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(Answer: $\binom{7}{3,2,2} = 210$)

Combination

Here is another way of thinking:

- Select 3 students from 7 students first for Room I. Can we figure out the number of such selections?
- Then select 2 students from the remaining 4 students for Room II. (6 selections)
- Then select 2 students from the remaining 2 students for Room III. (1 selection)

A selection of r objects from n without regard to order is a combination.

- Order does not matter.

Theorem:

The number of combinations of n distinct objects, taken r at a time is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

So we have another solution

$$\binom{7}{3}\binom{7-3}{2}\binom{7-3-2}{2} = \binom{7}{3}\binom{4}{2}\binom{2}{2} = \frac{7!}{3!2!\cdots 2!} = 210.$$

Summary:

How many ways can we choose r objects from n objects? To answer this question,

We need to know

(1) $\underline{\underline{\text{does order matter}}}$? and

(2) $\underline{\underline{\text{are we allowed to duplicate the objects}}}$?

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Example: 123, 213 are two different permutations of numbers $1, 2, 3$ but they have the same combination.

For a set of size n and a sample of size r, there are n^r different ordered samples with replacement and ${}_nP_r$ different ordered samples without replacement.

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Theorem:

The number of orderings of n elements is $n! = n(n-1)(n-2)\cdots 1$.

The number of unordered samples of r objects selected from n objects without replacement is

$$\binom{n}{r} = \frac{n!}{(n-r)!r!}$$

They are called the **binomial coefficients** as they are the coefficients in the expansion of

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}.$$

In particular, $2^n = \sum_{k=0}^{n} \binom{n}{k}$.

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For combinations with replacements, this is not so straightforward.

Sampling Table

Choosing r objects from a set of n objects.

	order matters	order doesn't matter
with replacement		
without replacement		

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	order matters	order doesn't matter
with replacement	n^r	$\binom{n+r-1}{r}$
without replacement	${}_nP_r$	$\binom{n}{r}$

See you next week.

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Tutorials will start next week